

FAIZA IQBAL

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EDUCATION

Master of Science (M.Sc.) in Data Science

Mehran University of Engineering and Technology

CGPA: 3.94/4.00

Thesis: An Explainable AI Approach to Speech-Based Alzheimer's Dementia Screening

Skills: Explainable AI, Machine Learning, Natural Language Processing

Feb 2023 - May 2025

Jamshoro, Pakistan

Bachelor of Engineering (B.E) in Telecommunication

Mehran University of Engineering and Technology

CGPA: 3.82/4.00

Thesis: Evaluating the Feasibility of Implementing Homomorphic Encryption

Skills: Cyber Security, Computer Networking, Machine Learning

Oct 2018 - Dec 2022

Jamshoro, Pakistan

WORK EXPERIENCE

Lecturer, Department of Computer Science

SZABIST University

August 2025 – Present

Hyderabad, Pakistan

- Teaching courses such as Artificial Intelligence, Programming for AI, Artificial Neural Networks and Natural Language Processing.
- Conducted various workshops related to AI and ML.
- Supervising multiple Final Year Projects in the domain of AI and ML.

Research Assistant - RADAR (Record Access and Data Retrieval Center)

Mehran University of Engineering and Technology

Nov 2023 - July 2025

Jamshoro, Pakistan

- Automated data pipelines using Python and SQL for large-scale publication and ranking datasets.
- Applied machine learning for trend analysis and predictive insights.
- Developed interactive dashboards and web tools to support research driven decision-making.

Research Assistant - Examination Department

Mehran University of Engineering and Technology

May 2023 - Nov 2023

Jamshoro, Pakistan

- Led data analysis on academic performance using SQL and Python.
- Designed predictive models to support institutional assessment strategies.
- Coordinated cross-departmental reporting as QOBE focal person.

MENTORING AND OUTREACH

HCIA AI Course Instructor

July 2026 - Present

- Delivering Huawei's Certified ICT Associate - Artificial Intelligence curriculum.

Mentor, VIP Program - Mental Health and Wellbeing Project

British Council

March 2025 – Present

- Mentored participants on a mental health and wellbeing focused project under the British Council's VIP programme.
- Presented project outcomes as a poster.

PUBLICATIONS

Aslam, A. B., Iqbal, F., Talpur, U., Syed, Z. S., & Shaikh, F. K. (2024). Artificial intelligence-enabled 6G mobile systems. In A. Bourdena et al. (Eds.), *Intelligent technologies for healthcare business applications*, pp. 49-79, Springer. https://doi.org/10.1007/978-3-031-58527-2_3

Iqbal, F., Syed, Z. S., Syed, M. S. S., and Syed, A. S., “An Explainable AI Approach to Speech-Based Alzheimer’s Dementia Screening,” in *Speech, Music and Mind 2024 (SMM24), Satellite Workshop of Interspeech 2024, International Speech Communication Association (ISCA)*, 2024, pp. 11-15. <http://dx.doi.org/10.21437/SMM.2024-3>.

SKILLS

Programming Languages: Python, MATLAB, C++, SQL
Libraries: Numpy, Pandas, Matplotlib, NLTK, SpaCy, Scikit-Learn, Keras, TensorFlow, PyTorch, LIME, SHAP
Tools and Technologies: Data Science, Machine Learning, Deep Learning, Feature Engineering, SQL Server, Latex, PowerBI, Git

PROJECTS

An Explainable AI Approach to Speech-Based Alzheimer’s Dementia Screening Developed interpretable machine learning models using linguistic and paralinguistic features, applying LIME and SHAP for transparent decision-making, which inspired my interest in model simplification and inherently interpretable architectures.
Tools and Techniques: Machine Learning, LIME, SHAP, Linguistic Feature Extraction, eGeMAPS, RMSE

NLP-Based Classification and Severity Assessment of Anxiety and Mental Health Conditions Designed an NLP system to classify anxiety severity and visualize psychological patterns through topic modeling and latent linguistic analysis.
Tools and Techniques: NLP, Topic Modeling, Word Clouds, Speech Analysis, Binary & Multi-Class Classification

Social Media Sentiment and Language Analysis Conducted NLP-based sentiment analysis on social media text data and extracted linguistic trends using visualization and feature-based modeling by applying word embedding and topic clustering methods
Tools and Techniques: Sentiment Analysis, NLP, Data Visualization, Word Clouds, Twitter Data Analytics

ACHIEVEMENTS

- **Postgraduate First Position Holder:** Achieved the highest CGPA in the MSc Data Science program.
- **Huawei ICT Competition Finalist (2022-2023):** Selected as one of the top 30 students in Pakistan in the prestigious Huawei ICT competition.
- **Top-10 Undergraduate Student:** Consistently ranked among the top-performing students during the Bachelor of Engineering program.
- **Academic Excellence Scholarship (2019-2022):** Awarded annual scholarships for three consecutive years in recognition of ranking among the top 10 students in the batch.

LANGUAGES

English (CEFR C1) German (CEFR B1)

RESEARCH CONFERENCES SYMPOSIUMS

- **18th Cyprus Workshop on Signal Processing and Informatics (CWSPI 2026)** Presented Abstract on ”A Comprehensive XAI Framework with Handcrafted Speech Features for Alzheimer’s Dementia Screening”
- **SMM Workshop, Interspeech 2024 – Presenter** Presented paper titled ”An Explainable AI Approach to Speech-Based Alzheimer’s Dementia Screening”, published by ISCA.

REFERENCES

Zafi Sherhan Syed Professor Mehran University of Engineering and Technology zafi.sherhan@faculty.muet.edu.pk	Sanam Narejo Associate Professor Mehran University of Engineering and Technology sanam.narejo@faculty.muet.edu.pk
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